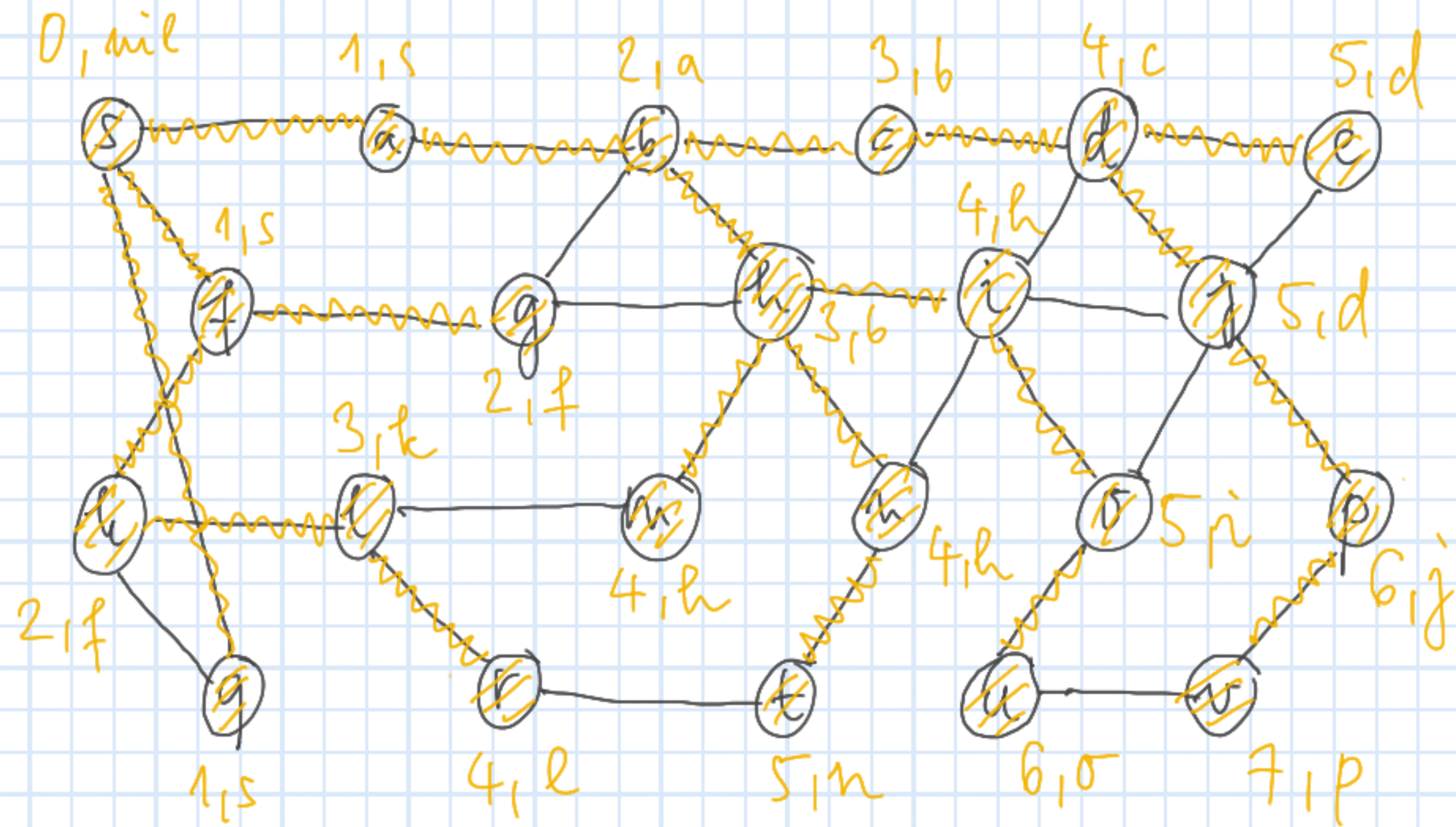


October 8, 2021

BFS sample execution



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a | f | q

f | q | b

q | b | g | e

b | g | k

g | k | c | h

k | c | h

c | h | e

h | e | d

e | d | i | m | m

d | i | m | m | r

i | m | h | r | e | j

m | h | r | e | j | o

m | r | e | j | o

r | e | j | o | t

e | j | o | t

j | o | t

o | t | p

t | p | m

p | m

m | r

r

∅

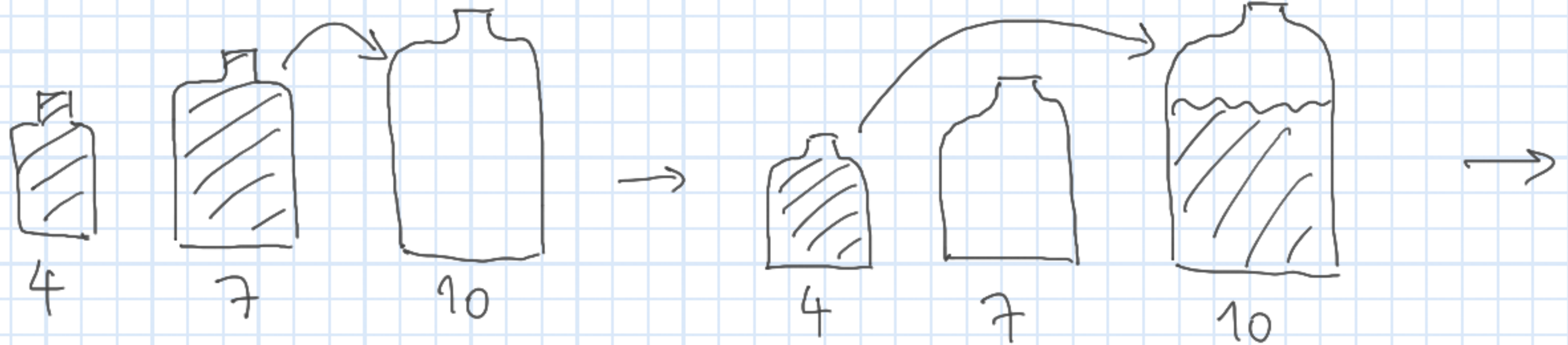
Die Hard 3 variant

We have 3 jugs of capacity 4 gallons, 7 gallons, 10 gallons, resp.

Initially the two smaller jugs are full of water, the biggest one is empty.

The goal is to measure exactly 2 gallons of water in one of the two smaller jugs.

You are allowed to pour water from one jug to another but you must stop pouring when the first jug becomes empty OR the second one becomes full





4



7



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Is it possible? If yes, what is the
smallest number of steps (pourings)
to get 2 gallons of water?

You can solve both problems (decision, min. number of steps) using BFS on an appropriate graph:

STATE GRAPH

vertices $\rightarrow$ states

which ing how filled

edges (directed) $\rightarrow$ one state can be obtained from another one by one pouring

Problem

Find a shortest path from the initial state to a "desired" one

States

$(a, b, c) \rightarrow$ amount of water
in gallons in the 4-gallon, 7-gallon,
10-gallon jug | HSP.

$$0 \leq a \leq 4$$

$$0 \leq b \leq 7$$

$$0 \leq c \leq 10$$

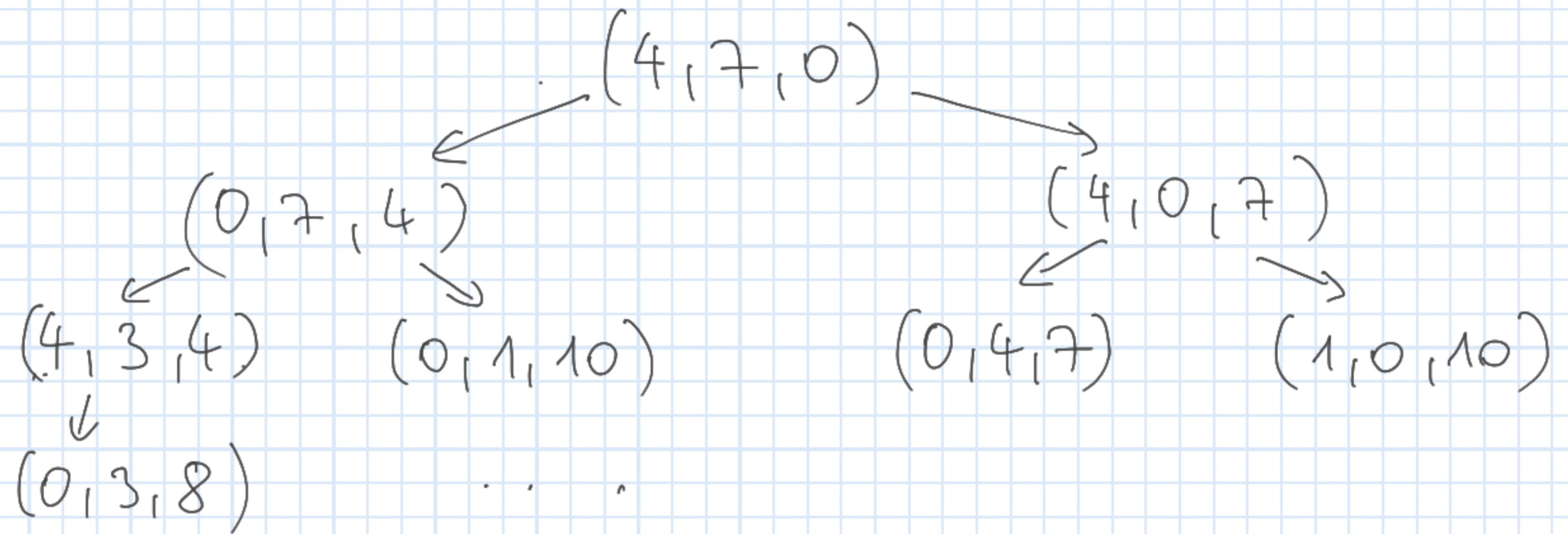
$$a + b + c = 11$$

$$a, b, c \in \mathbb{N} \quad (\text{integers})$$

How can we run BFS on this state graph?

→ Build the state graph and run BFS

→ build the state graph ^(the BFS tree, in fact) simultaneously
with the running of BFS ☺



HW. Reach $(*, 2, *)$ or $(2, *, *)$!

